Chandrakanth Gudavalli

3D Computer Vision Engineer
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Professional Summary

3D Computer Vision Engineer specializing in multi-view geometry, camera pose estimation, large scale 3D reconstruction, and neural scene representations including 3D Gaussian Splatting and NeRFs. Experienced in building scalable computer vision systems for real-world imagery, spanning research prototyping through production deployment. Strong background in Structure-from-Motion, calibration, localization, and scene understanding, with hands-on expertise developing robust and reliable 3D vision pipelines.

Work Experience

Research Engineer, 3D Computer Vision **(December 2019 - Present)**
Mayachitra Inc (Santa Barbara, California)

- Productionization of Cloud-native 3D reconstruction & camera estimation pipelines (IARPA WRIVA): Designed and productionized a cloud-ready Structure-from-Motion pipeline robust to wide variations in altitude, viewpoint, and illumination. The system integrates GPS metadata, OpenStreetMap priors, and satellite imagery to generate scalable sparse and dense reconstructions, serving as the foundation for downstream 3DGS and NeRF pipelines. The deployed system is consistently ranked #1 on the IARPA WRIVA leaderboard, outperforming teams from MIT and Stanford. [Sample Demo]
- End-to-end Ground to Satellite Geolocalization Pipeline: Built an end-to-end novel view synthesis and geolocalization pipeline that aggregates information from clusters of ground-level images into a 3DGS–based zenith render using Structure from Motion, and post process the 3DGS render with Difix3D, enabling geolocation of cameras and objects relative to geo-registered satellite imagery from WorldView-3. Owned the workflow from data ingestion through rendering, evaluation, and metric reporting, achieving sub-10 m RMSE in camera GPS estimation. [CVPR *2026* Paper]
- Custom PnP Solvers to enable large scale 3D reconstruction by leveraging Noisy GPS: Designed a custom PnP solver to add a new camera to an existing 3D point cloud by leveraging the noisy GPS metadata associated with the image and the spatial neighbors of the image. This resulted in a module that can add an image to the existing point cloud in less than 1.34 seconds. [ICIP 2025 – MetaReg]
- License-Free "Feed-Forward" Camera Parameter Estimation Pipeline: Conducted in-depth evaluation of feed-forward camera parameter estimation models (e.g., VGGT, Pi3, DAv3), analyzing their strengths and limitations relative to optimization-based SfM pipelines such as COLMAP, GLOMAP, and InstantSfM. Developed rapid prototyping workflows to benchmark pose estimation quality across methods and designed pipelines capable of estimating reliable relative camera poses even from extremely small image sets (as few as 2–5 images).
- Semantics aware generative image generation pipelines: Developed a GAN-based image manipulation framework supporting object insertion and removal in aerial and consumer imagery using few-shot learning and semantic priors. Designed the system with explicit training and inference stages, reproducible configurations, and well-defined outputs suitable for downstream evaluation and forensic analysis. [Publication link] · [Approved Patent]
- Physics-based metadata authentication pipelines for satellite imagery: Designed physics-driven verification pipelines to authenticate Rational Polynomial Coefficient (RPC) metadata used in satellite image

orthorectification. The system achieves 99.98% accuracy by modeling and matching DFT patterns of pixel displacement, and has been validated through both peer-reviewed publication and an approved U.S. patent. [Publication Link] [Approved Patent]

Visiting Research Staff Member (ML Ops & 3D Vision) (Part Time) **(June 2022 - Present)**
Vision Research Laboratory, UCSB

- Owned and led end-to-end development of 3D Reconstruction systems, reconstructing campus-scale environments from heterogeneous, sparse imagery. Designed and operated the full lifecycle spanning data ingestion, camera pose estimation, reconstruction, validation, and reproducible artifact generation. Scaled systems to thousands of images while maintaining stability, predictable performance, and debuggability. [Sample Demo-1] · [Sample Demo-2]
- Technical lead for a cross-institution collaboration with Johns Hopkins University, coordinating algorithm development, infrastructure design, and large-scale evaluations across a 14-member research team, with responsibility for aligning research direction to deployable system constraints.
- Productionized the BisQue 3D vision platform, transforming it from a research-oriented environment into a containerized, *Kubernetes-backed system capable of serving large-scale 3D and 4D vision workloads*. Redesigned architecture so computer vision and graphics pipelines deploy as modular services with explicit inputs, outputs, and GPU resource contracts.
- Designed and operated backend services and orchestration layers for ML workloads, enabling GPU-backed execution via Kubernetes scheduling and Argo workflows. Implemented robust job submission, monitoring, and failure recovery mechanisms, clearly separating *user-facing requests, compute execution, and generated artifacts*, significantly improving reliability under concurrent workloads.
- Developed and deployed user-facing 3D visualization and interaction pipelines, enabling real-time inspection of volumetric and point-based representations (CT volumes, point clouds, 3D Gaussian Splats). Focused on exposing complex 3D outputs through responsive, production-ready interfaces suitable for downstream applications.

Graduate Intern, Computer Vision **(March 2019 - December 2019)**
National Renewable Energy Laboratory (Denver, Colorado)

- Project 1 – Image segmentation model for receiver edge detection: Developed a semantic segmentation model with a modified SegNet architecture to detect the reflection of the solar tower on heliostats. Shape of the receiver's reflection on the heliostat is used for picking defective facets. SegNet Accuracy: 92.14%
- Project 2 – Anomaly Detection in Bio-Feedstock stream: Developed a GRU network based time series classification model to detect if the input feedstock stream can overload the reactor and lead to unnecessary shutdown. Publication Link: https://rdcu.be/caZd9

Software Engineer **(June 2015 – July 2017)**
Netcracker Technologies (Hyderabad, India)

- Worked for 2 years as C++ and Python developer of telecom billing applications (ICOMS, RB).
- Worked towards the development of REST APIs to streamline customer payments and adjustments that are compliant with PCI standards.

Education

M.S. Electrical and Computer Engineering **(August 2017 – December 2019)**
CGPA: 3.9
Major: Computer Vision and Embedded Systems
University of Colorado, Denver
Related Courses: Computer Vision, Neural Networks for Computer Vision, Machine Learning, Information Theory, Random Processes for Engineering

Bachelor of Technology, Electronics and Communication Engineering **(April 2011 – May 2015)**
CGPA: 3.6
Jawaharlal Nehru Technological University, India
Related Courses: Digital Image Processing, Algorithms and Data Structures, Probability Theory, Applied Statistics, Engineering Mathematics, Signals and Systems

Relevant Publications (Papers and Patents)

#	Year	Title and Authors	Publisher
1	2025	C. Gudavalli et al., "Towards Spatial Intelligence for Geo-locating Ground Images onto Satellite Imagery"	CVPR 2026 (Main Conference)
2	2025	C. Gudavalli et al., "METAREG: Robust Camera Parameter Estimation by Leveraging Noisy Camera Extrinsics"	IEEE International Conference on Image Processing (ICIP)
3	2025	Gudavalli, Chandrakanth, Erik Rosten, Lakshmanan Nataraj, Shivkumar Chandrasekaran, and B. S. Manjunath. "CIMGEN: Controlled Satellite Image Manipulation by Fine Tuning Pretrained Generative Models on Limited Data."	International Conference on Pattern Recognition, pp. 95-110. Springer, Cham, 2025.
5	2024	Devendra K Jangid, Neal R Brodnik, McLean P Echlin, Chandrakanth Gudavalli, Connor Levenson, Tresa M Pollock, Samantha H Daly, BS Manjunath “Q-RBSA: high-resolution 3D EBSD map generation using an efficient quaternion transformer network”	Nature NPJ Computational Materials
6	2023	Chandrakanth Gudavalli, Michael Goebel, Tejaswi Nanjundaswamy, Lakshmanan Nataraj, Shivkumar Chandrasekaran, BS Manjunath “Resampling Estimation based RPC Metadata Verification in Satellite Imagery”	IEEE Transactions on Information Forensics and Security
7	2023	Chandrakanth Gudavalli, Michael Goebel, Tejaswi Nanjundaswamy, Lakshmanan Nataraj, Shivkumar Chandrasekaran, Bangalore S Manjunath “RATIONAL POLYNOMIAL COEFFICIENT BASED METADATA VERIFICATION”	US Patent Office
8	2022	Gudavalli, Chandrakanth, Erik Rosten, Lakshmanan Nataraj, Shivkumar Chandrasekaran, and B. S. Manjunath. "SeeTheSeams: Localized Detection of Seam Carving based Image Forgery in Satellite Imagery."	CVPR 2022 - Workshop on Media Forensics
9	2021	Nataraj, Lakshmanan, Chandrakanth Gudavalli, Tajuddin Manhar Mohammed, Shivkumar Chandrasekaran, and B. S. Manjunath. "Seam Carving Detection and Localization Using Two-Stage Deep Neural Networks."	Machine Learning, Deep Learning and Computational Intelligence for Wireless Communication Springer, Singapore, 2021.
10	2020	Gudavalli, Chandrakanth, Elizabeth Bose, Bryon S. Donohoe, and David A. Sievers. "Real-time biomass feedstock particle quality detection using image analysis and machine vision."	Biomass Conversion and Biorefinery (2020)

Checkout my website for more details: www.chandrakanthgudavalli.com